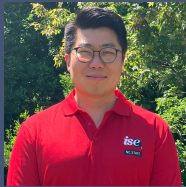


Donghyun Ko



PROFILE

Hi, I am a Ph.D. student in the ISE department at NC state university, focusing on risk-aware system control and optimization using ML/DL and Generative AI. My research centers on AI modeling for uncertainty-aware predictive analytics and manufacturing process optimization. Complements academic research with 7+ years of hands-on OR and ML deployment in high-stakes defense environments.

CONTACT DETAILS

- ✉ dko3@ncsu.edu
- ☎ 919-454-2479
- 🌐 Webpage: kmakdh3692.github.io
- 🌐 linkedin.com/in/donghyun-ko
- 🐙 github.com/kmakdh3692
- 🆔 orcid.org/0000-0003-2182-5733

AREAS OF INTEREST

- Generative AI
- Uncertainty Quantification
- Machine/Deep Learning
- RUL/TTF Prediction
- Anomaly Detection
- AM process control and parameter optimization
- Defense OR & Simulation

TECHNICAL SKILLS

- **ML & DL modeling**
- **Optimization:** Linear&Nonlinear, Integer, Stochastic Programming
- **Programming:** Python (PyTorch, TensorFlow, scikit-learn), R, SQL
- **Analytics:** Stochastic Process Control; Design of Experiment; Response Surface Methods

LANGUAGES

Korean (native)
English (professional)

EDUCATION * DETAILED COURSES TAKEN AVAILABLE IN APPENDIX

- North Carolina State University, Raleigh, NC** **08/2023–Present**
- ◇ Ph.D. student in Industrial & Systems Engineering (GPA: 3.91 / 4.0)
 - ◇ Research focus: Risk-aware Artificial Intelligence modeling for industrial predictive analytics and additive manufacturing process control & optimization
 - ◇ Advisor: Prof. Jake Peloquin
 - ◇ Key coursework: Machine & Deep learning, Generative AI, Inverse problem, Active learning optimization, Stochastic and Bayesian modeling
- Texas A&M University, College Station, TX** **08/2019–05/2021**
- ◇ M.S. in Industrial & Systems Engineering (GPA: 3.9 / 4.0)
 - ◇ Thesis: Nonlinear system optimization via response surface methodology with the case study of maximizing UAV flight time
 - ◇ Key coursework: Optimization, Stochastic processes, Design of experiment, Statistical computing, Machine & Deep learning
- Korea Military Academy, Seoul, South Korea** **03/2008–02/2012**
- ◇ B.A. in Economics (GPA: 3.96 / 4.3)
 - ◇ Emphasis on quantitative analysis and mathematical & engineering fundamentals

RESEARCH EXPERIENCE

- Research Assistant — CAMAL Lab, NC State** **08/2025–Present**
Advisor: Prof. Jake Peloquin | Raleigh, NC
- ◇ Developing a conditional diffusion model for risk-aware process control and parameter optimization in metal additive manufacturing
 - ◇ Manuscript in preparation: “Risk-Aware Process Control for Metal Additive Manufacturing via Conditional Diffusion Model” (Ko & Peloquin, 2026)
- Research Assistant — QCRE Lab, NC State** **08/2024–Present**
Advisor: Prof. Xiaolei Fang | Raleigh, NC
- ◇ Developing a conditional diffusion model as a robust tool for predicting time-to-failure of engineering systems under challenging and realistic scenarios
 - ◇ Manuscript under review: “A Conditional Diffusion-Based Generative AI Model for Industrial Predictive Analytics.” (Ko & Fang, 2026)
- Operations Research & Analytics Officer** **06/2022–06/2023**
Center for Army Analysis & Simulation, Army HQ, South Korea
- ◇ Designed time-series forecasting models for personnel resource demand and training facility utilization, directly informing national-level capacity planning
 - ◇ Formulated an integer programming model to optimize Recruit Training Center capacity allocation, supporting long-range enlisted personnel planning
 - ◇ Re-engineered recruit training workflows (layouts & schedules), identifying bottlenecks and reducing operational budget by 15%
- Maneuver Logic Simulation & Analytics Officer** **05/2021–06/2022**
Battle Command Training Program, Army HQ, South Korea
- ◇ Built an ML/DL-based AI decision-support prototype for the Battle Command Training Program, cutting manual planning cycles for war-game simulations
 - ◇ Optimized deployment of 105 mm self-propelled artillery assets along the DMZ using cost-effectiveness analysis, improving readiness within fixed budget constraints
 - ◇ Developed and maintained algorithms for military war-gaming simulations

TEACHING EXPERIENCE

- Teaching Assistant, NC State University**
- ◇ TA, ISE 443 – Quality Control & Design (Fall 2025)
 - ◇ TA, ISE 311 – Engineering Economic Analysis (Spring 2025)
 - ◇ TA, ISE 135 – Computer-Based Modeling for Engineers (Spring & Fall 2024)
 - ◇ TA, ISE 535 – Python for Industrial & Systems Engineering (Fall 2023)

PUBLICATIONS

[Under Review] Donghyun Ko, Xiaolei Fang (2026).

"A Conditional Diffusion-Based Generative AI Model for Industrial Predictive Analytics." Under review at EAAI. Preprint (SSRN): <https://doi.org/10.2139/ssrn.6169178>

[In Progress] Donghyun Ko, Jacob Peloquin (2026).

"Risk-Aware Process Control & Parameter Optimization for Metal Additive Manufacturing via Conditional Diffusion Model with Co-Registered Digital-Twin Data."

[Published] Donghyun Ko, Juneyoung Bang (2022).

"Application of Response Surface Method on Non-linear System Optimization: Maximization of military microdrone's flight time." *Korean Management Science Review*, 39(1), 1–14. <https://doi.org/10.7737/kmsr.2022.39.1.001>

[Published & Classified] Donghyun Ko (2022).

"Building a model for predicting the success rate of a military operation with the use of a variety of ML algorithms: Focusing on the phase of counter-attack." *Army Future Innovation & Technology* (36-981AGG-010534-06): 87–116.

TALKS & OTHERS

[Presentation] "Risk-Aware Process Control of Additive Manufacturing via Conditional Diffusion Model." *2026 IISE Annual Conference & Expo*, Arlington, TX, May 2026.

[Presentation] "A Conditional Diffusion-based Generative AI Model for Industrial Predictive Analytics." *2026 IISE Annual Conference*, Arlington, TX, May 2026.

[3-Minute Speech] "Doctoral Colloquium Discussions & 3-minute thesis competition" *2026 IISE Annual Conference & Expo*, Arlington, TX, May 2026.

[Presentation] "IISE QCRE track best paper competition" *2026 IISE Annual Conference & Expo*, Arlington, TX, May 2026.

[Poster] Additive Manufacturing Process Parameter Optimization by Generative AI. *NC State Graduate Research Symposium*, April 2026.

[Presentation] "Diffusion Models for Robust RUL/TTF Prediction under Complex Degradation." *AI4SE & SE4AI Workshop*, Washington, DC, Sep 2025. (Included in official workshop report)

[Presentation] "Response Surface Methodology for Nonlinear System Optimization." *International Army Modeling & Simulation Conference*, Daejeon, Korea, Nov 2023.

HONORS

[Winner] Doctoral Colloquium 3-Minute Thesis Competition. *2026 IISE Annual Conference*, Arlington, TX, May 2026. (News)

[Finalist] IISE QCRE Best Track Paper Competition. *2026 IISE Annual Conference*, Arlington, TX, May 2026. (News)

A member of Institute of Industrial and Systems Engineers (IISE)

CERTIFICATIONS

Big Data Analyst Engineer

Korea Data Agency (K-DATA)

◇ Government-certified credential validating end-to-end data analytics, statistics, and machine learning proficiency

Advanced Data Analytics Professional (ADP)

K-DATA

◇ Highest-level data analytics certification in Korea. Rigorous evaluation of advanced statistical modelling, optimization, and ML/DL expertise

Advanced Data Analytics Semi-Professional (ADsP)

K-DATA

◇ Professional certification confirming strong foundations in data analysis and statistical inference

NVIDIA Certificate of Competency — Deep Learning

NVIDIA

- ◇ Industry-recognized certification demonstrating practical proficiency in deep learning fundamentals and GPU-accelerated AI workflows

INTERNSHIP & TRAINING

Interpreter, 22nd ROK Armed Forces Anniversary Event **09–10/2022**

- ◇ Simultaneous interpretation for senior military officials, including U.S. 7th Air Force Commander, U.S. 2nd Infantry Division Commander, and the Swedish Ambassador.

Army Operations Course, Canadian Armed Forces, Kingston **06–12/2017**

- ◇ Completed brigade–division level staff officer course with certification in operational planning and joint military operations.

Defense Language Institute, Republic of Korea **09/2014–02/2015**

- ◇ Completed intensive English language training program for international military and defense operations.

LEADERSHIP

Company Commander, ROK Army (2017–2019)

- ◇ Led company-level operations and training; ensured combat readiness, troop management, and unit discipline
- ◇ Conducted high-security missions including protection of the US–ROK Combined Command facility and presidential security operations

Division Logistics Officer (S-4), ROK Army (2016–2017)

- ◇ Forecasted combat resource requirements and coordinated replenishment, maintenance, and combat service supports

Brigade Operations Planning Officer (S-3), ROK Army (2015–2016)

- ◇ Developed brigade-level infantry operational plans, conducted risk assessments, and established rules of engagement

Battalion Operations Planning Officer (S-3), ROK Army (2013–2014)

- ◇ Developed battalion-level infantry operational plans, conducted risk assessments, and established rules of engagement for border security and DMZ operations

Platoon Leader (2012–2013)

- ◇ Commanded platoon-level units responsible for border security, readiness, training, and rapid response under established rules of engagement

APPENDIX: DETAILED COURSEWORK

North Carolina State University **Ph.D.**

- ◇ Linear Programming (ISE 505); Statistical Models for System Analytics in ISE (ISE 537); Applied Stochastic Models (ISE 760); High-Dimensional Data Analysis (ISE 789); Manufacturing Processes (ISE 515); Human Factors System Design (ISE 540); Practical Machine Learning for Engineering (ISE 538); Fundamentals of Statistical Inference I–II (ST 501/502); Experimental Statistics for Engineers I–II (ST 515/516); Applied Bayesian Analysis (ST 540); Neural Networks (ECE 542); Algorithmic Methods in Nonlinear Programming (OR 506); Scientific Machine Learning (NE 795).

Texas A&M University **M.S.**

- ◇ Linear Programming; Nonlinear Programming; Distribution Theory; Stochastic Processes; Design of Experiments; Advanced Quality Control; Statistical Computing with R and Python; Engineering Data Analysis; Data Mining; Applied Numerical Analysis.

Korea Military Academy **B.A.**

- ◇ Introduction to Computers; Computer Networks; Computer Programming; Linear Algebra; Probability & Statistics; Calculus; Information Warfare; Fundamental Physics; Fundamental Chemistry; Introduction to Civil Engineering; Introduction to Mechanical Engineering; Weapon Systems Engineering; Social Science Research Methods (SPSS); Economics (Micro, Macro, Econometrics, International, Public, Industrial Organization); Cost & Benefit Analysis; Wargame Theory; National Defense Economics; etc.